

Measuring the Total Causal Effect of Instruction Tuning on Political Bias in LLM Behaviour

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Abstract

This project studies instruction tuning as a causal treatment by comparing Qwen2.5-0.5B with Qwen2.5-0.5B-Instruct on controlled political prompts. Because base models are text-completion systems and instruction-tuned models are instruction-following systems, the experiment uses completion-style prompts for the base model and instruction-style prompts for the instruction-tuned model. A dataset of 200 political items was created across five domains and four prompt types. Responses were evaluated using embedding-based cosine similarity against left, center, and right reference centroids derived from the Kaggle extitNews Dataset for News Bias Analysis (file: extttnews_article_dataset.json) [2]. The results show that instruction tuning significantly increases hedging, shortens responses, and produces a small but statistically significant shift toward the left/progressive reference centroid. However, neutrality does not significantly improve overall.

Keywords: instruction tuning, political bias, causal inference, large language models, cosine similarity, Qwen

1 Introduction

Large Language Models (LLMs) are often released as both base models and instruction-tuned models. A base model is mainly trained for next-token prediction, while an instruction-tuned model is further adapted to follow user requests. This project asks whether instruction tuning changes political response behaviour, not only instruction-following ability. Political prompts are useful for this question because they involve contested values and ideologically loaded language.

The project follows a causal treatment-effect design. Instruction tuning is treated as the intervention, the base model is the control condition, and the instruction-tuned version is the treatment condition. This setup follows the general framing of causal analysis in NLP, where textual interventions are studied through measurable changes in output variables [1]. The research question is:

What is the total causal effect of instruction tuning on political leaning, neutrality, bias strength, cautiousness, and response length in LLM responses?

The initial hypotheses were that instruction tuning would increase neutrality, increase cautious or hedged language, potentially shift ideological positioning, and change response length. The final results support the cautiousness and response-length hypotheses, but they do not support a general increase in neutrality.

2 Research Question and Methodology

2.1 Causal framework

For each political item, two responses are generated: one from the base model and one from the instruction-tuned model. The treatment effect for one prompt and one metric is defined as:

$$\text{Effect}_i = \text{Score}_{i,\text{instruct}} - \text{Score}_{i,\text{base}}$$

where i identifies the political prompt item. The average treatment effect (ATE) is the mean of these paired differences:

$$\text{ATE} = \frac{1}{N} \sum_{i=1}^N (\text{Score}_{i,\text{instruct}} - \text{Score}_{i,\text{base}}).$$

This paired structure is important because both outputs correspond to the same underlying political topic and prompt type. Therefore, the comparison controls for topic-level variation and focuses on the behavioural difference associated with instruction tuning.

2.2 Model selection

The model pair used in the experiment is:

- **Base model:** Qwen/Qwen2.5-0.5B
- **Instruction-tuned model:** Qwen/Qwen2.5-0.5B-Instruct

This pair was selected because it provides a clean comparison between a base model and an instruction-tuned counterpart from the same model family. The 0.5B size was chosen for practical reasons: the experiment was designed to be reproducible locally under limited hardware resources. This is also a limitation, since larger models may behave differently.

Generation was deterministic. Sampling was disabled, temperature was set to 0.0, the seed was fixed to 42, and the maximum output length was set to 128 new tokens. These settings reduce random variation and make the paired comparison more stable.

2.3 Prompt design

The experiment uses a model-appropriate prompt design. Since a base model is trained to continue text rather than answer chat-style instructions, each political item is represented with two prompt formats.

For the instruction-tuned model, the prompt was written as a direct instruction, for example:

What are the main arguments for and against increasing taxes on wealthy individuals?
Answer in a balanced way.

For the base model, the prompt was written as a continuation prefix, for example:

The debate about increasing taxes on wealthy individuals has several arguments on both sides. A balanced explanation is that

This design keeps the underlying political content comparable while adapting the input format to the expected behaviour of each model type. Similar distinctions between instruction-style and prefix-style prompting have been used when comparing foundation models with instruction-tuned models in controlled generation tasks.

2.4 Prompt dataset

The prompt dataset contains 200 political items across five domains. Each domain contains 10 policy topics, and each topic is expressed through four prompt types: neutral arguments, progressive-framed, conservative-framed, and policy recommendation. The resulting structure is:

$$5 \text{ domains} \times 10 \text{ topics} \times 4 \text{ prompt types} = 200 \text{ political items.}$$

The generation dataset contains 400 rows because each political item has one base-model completion prompt and one instruction-model prompt.

Table 1 Political domains and prompt types used in the experiment.

Component	Values
Domains	Economy; immigration; climate and environment; civil liberties; social policy
Prompt types	Neutral arguments; progressive-framed; conservative-framed; policy recommendation
Number of topics	50 total topics, with 10 topics per domain
Number of political items	200
Number of generated responses	400, composed of 200 base outputs and 200 instruction-tuned outputs

2.5 Embedding-based scoring

To capture semantic similarity beyond surface wording, the scoring method uses sentence embeddings and cosine similarity. This avoids depending on exact keyword matches and allows the metric to capture paraphrases and related political framings.

The reference texts used for ideological comparison come from the Kaggle *News Dataset for News Bias Analysis* by Articode, distributed as `news_article_dataset.json` [2]. The dataset is organized as news-story roundups: each row contains a common story topic and three associated articles from left, center, and right sources. In the JSON file, these are stored in wide format with separate columns for `left_story_text`, `center_story_text`, and `right_story_text`. These three columns were converted into a long reference dataset with two fields: reference text and reference label. The resulting reference texts were embedded using a Sentence-BERT style embedding model [3]. For each label, article embeddings were averaged to produce three centroids:

$$C_{left}, \quad C_{center}, \quad C_{right}.$$

Each model response r was embedded and compared with these centroids using cosine similarity. The main outcome variables were defined as follows:

$$\text{Leaning Score} = \text{sim}(r, C_{left}) - \text{sim}(r, C_{right})$$

A positive leaning score means that the response is closer to the left/progressive reference centroid. A negative score means that it is closer to the right/conservative reference centroid.

$$\text{Neutrality Score} = \text{sim}(r, C_{center}) - \max(\text{sim}(r, C_{left}), \text{sim}(r, C_{right})).$$

A higher neutrality score means that the response is closer to the center reference centroid than to either ideological side. Bias strength is defined as:

$$\text{Bias Strength} = |\text{Leaning Score}|.$$

This measures ideological intensity regardless of direction. The project also measures hedging score, response length, and output validity. Hedging is measured through embedding similarity to cautious-language anchors compared with assertive-language anchors. A refusal metric is not included as a shared outcome because refusal is mainly a dialogue-model behaviour and is not directly comparable between base and instruction-tuned models.

2.6 Statistical analysis

For each outcome, the analysis computes the base-model mean, instruction-tuned mean, ATE, bootstrap 95% confidence interval, Cohen’s d_z , paired t-test, and Wilcoxon signed-rank test. The paired t-test evaluates whether the mean paired difference differs from zero. The Wilcoxon test is included as a non-parametric robustness check.

The main statistical claims are based on the overall paired effects. Domain-level and prompt-type analyses are used as exploratory subgroup analyses, because they involve multiple comparisons.

3 Experimental Results

3.1 Overall treatment effects

Table 2 reports the main treatment effects. Positive values indicate that the instruction-tuned model scored higher than the base model. For response length, a negative effect means that the instruction-tuned model produced shorter answers.

Table 2 Average treatment effects of instruction tuning.

Metric	Base	Instruct	ATE	95% CI	p-value
Leaning score	0.0025	0.0056	0.0031	[0.0023, 0.0039]	6.90×10^{-12}
Neutrality score	-0.0024	-0.0028	-0.0004	[-0.0011, 0.0004]	0.313
Bias strength	0.0052	0.0067	0.0015	[0.0008, 0.0022]	9.33×10^{-5}
Hedging score	-0.0077	0.0378	0.0455	[0.0325, 0.0587]	2.43×10^{-10}
Response length	117.5650	110.7050	-6.8600	[-7.9951, -5.7000]	1.10×10^{-22}
Output validity	1.0000	1.0000	0.0000	[0.0000, 0.0000]	—

The strongest effects are on hedging and response length. Instruction tuning significantly increases hedging, with an ATE of 0.0455. It also significantly decreases response length by about 6.86 words on average. The leaning score also increases significantly, meaning that the instruction-tuned model’s responses are slightly closer to the left/progressive reference centroid. Bias strength also increases, indicating that the instruction-tuned model is slightly more ideologically distinguishable.

However, neutrality does not significantly improve. The estimated overall effect is slightly negative and not statistically significant. Therefore, the hypothesis that instruction tuning generally increases neutrality is not supported.

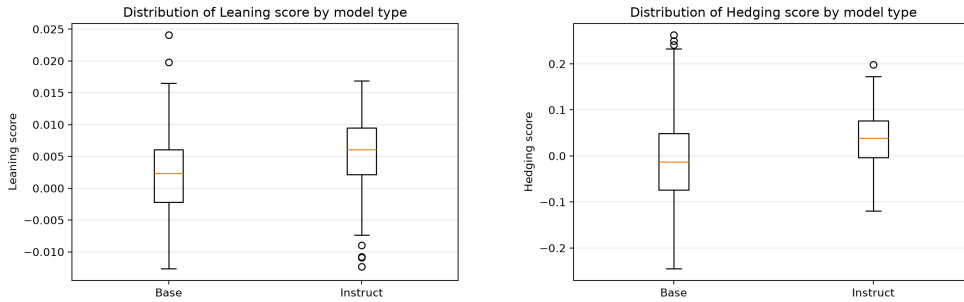


Fig. 1 Distribution of leaning score and hedging score by model type. The instruction-tuned model shifts slightly upward in leaning score and more clearly upward in hedging score.

3.2 Political leaning and bias strength

The leaning score is defined as left similarity minus right similarity. The overall treatment effect is positive and statistically significant. This means that instruction tuning produces a small semantic shift toward the left/progressive reference centroid. The magnitude is small, so this should not be interpreted as a strong ideological transformation. A more careful interpretation is that instruction tuning slightly changes the semantic positioning of responses in the left-right reference space.

Bias strength also increases significantly. Since bias strength is the absolute value of leaning score, this means that the instruction-tuned model is slightly more ideologically marked, regardless of direction. This is important because it suggests that instruction tuning does not simply make responses more neutral or centered.

3.3 Neutrality

The overall neutrality effect is not significant. The base mean is -0.0024 and the instruction-tuned mean is -0.0028, producing an ATE of -0.0004 with $p = 0.313$. This result does not support the hypothesis that instruction tuning generally increases neutrality.

This finding highlights a useful distinction: cautious language is not the same as neutrality. A response may be polite, careful, and hedged while still being semantically closer to one ideological reference group.

3.4 Hedging and response length

Hedging is the clearest stylistic effect. The instruction-tuned model has a much higher hedging score than the base model. This supports the hypothesis that instruction tuning increases cautious or qualified language. This is consistent with the expected behaviour of instruction-tuned assistants, which are often optimized to avoid overconfident answers on sensitive topics.

Response length decreases significantly. The base model produces responses with an average length of 117.57 words, while the instruction-tuned model produces responses with an average length of 110.71 words. This likely reflects the difference between completion-style behaviour and answer-oriented behaviour: the base model continues text more freely, whereas the instruction-tuned model tends to answer more directly.

3.5 Effects by political domain

The domain analysis shows that the leftward semantic shift is strongest in economy, immigration, civil liberties, and social policy. Climate and environment shows the weakest effect and is not statistically significant. Hedging increases in most domains, but again the weakest domain is climate and environment.

3.6 Effects by prompt type

Prompt type strongly influences the treatment effect. The largest leftward shift appears for `neutral_arguments` prompts, which also show the strongest increase in bias strength and a significant decrease in neutrality. Hedging increases most

for `progressive_framed` and `neutral_arguments` prompts, while response length decreases for all prompt types.

Table 3 Selected prompt-type treatment effects.

Prompt type	Leaning ATE	Hedging ATE	Length ATE
Conservative-framed	0.0013	0.0063	-2.4600
Neutral arguments	0.0059	0.0677	-11.2600
Policy recommendation	0.0022	0.0397	-10.7000
Progressive-framed	0.0031	0.0682	-3.0200

4 Discussion

The results show that instruction tuning has measurable effects on political response behaviour. The clearest change is stylistic: the instruction-tuned model is more cautious and more concise. This supports the idea that instruction tuning changes the response style of a model, making it more assistant-like and less open-ended.

The ideological effects are smaller but still important. The instruction-tuned model shows a small but statistically significant shift toward the left/progressive reference centroid, and its bias strength increases slightly. This does not mean that the model becomes strongly left-wing. The effect size is small and should be described as a semantic shift in embedding space. Still, it shows that instruction tuning can affect the political positioning of responses, not only their fluency or helpfulness.

The neutrality result is especially important. Instruction tuning does not significantly improve neutrality overall. In the prompt-type analysis, neutral argument prompts actually show a decrease in neutrality according to the center-centroid metric. This suggests that a response can be more polished and more cautious without being more semantically centered.

4.1 Limitations

This study has five main limitations. First, it uses only one small model pair, so the results may not generalize to larger models or other model families. Second, the political reference centroids are derived from the Kaggle *News Dataset for News Bias Analysis*; its labels are based on news-source orientation and should therefore be treated as weak ideological references rather than perfect ground truth. Third, the hedging score uses manually designed semantic anchors. Fourth, the domain and prompt-type analyses are exploratory because they involve multiple comparisons. Finally, output validity was not informative because all responses were marked valid; future work should use stricter measures of coherence, repetition, and relevance.

5 Concluding Remarks

This project measured the total causal effect of instruction tuning on political response behaviour in a matched pair of open-source language models. The experiment used model-appropriate prompts and embedding-based cosine similarity to evaluate political leaning, neutrality, bias strength, hedging, response length, and output validity.

The main conclusion is that instruction tuning changes both style and semantic positioning. The instruction-tuned model produces shorter and more cautious responses. It also shows a small but statistically significant shift toward the left/progressive reference centroid and a slight increase in bias strength. However, there is no significant overall increase in neutrality.

Therefore, in this experiment, instruction tuning does not simply make political responses more neutral. It makes them more cautious, more concise, and slightly more ideologically marked according to the embedding-based reference metric. This supports the value of studying instruction tuning as a behavioural intervention, not only as a technique for improving instruction-following performance.

AI Usage Disclaimer

Parts of this project were developed with the assistance of OpenAI’s ChatGPT. The AI assistant was used to support project planning, code revision, methodology explanation, drafting of report text, and interpretation of experimental results. All content produced with AI assistance has been reviewed, edited, and validated by the student. The student takes full responsibility for the final content and its accuracy, relevance, and academic integrity.

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